

The goal of Ch 3 is to work on optimization problems, these are problems in which we have a model/problem and want to find the optimal/best solution by finding some local min/max.

Some general examples:

- minimize cost / maximize Profit
- Maximum speed
- Maximum Area
- etc.

There are a lot of problems where the solution involves finding the maximum or minimum of a function.

Def: Let c be a number in the domain of f . Then $f(c)$ is the

↳ Sometimes max or mins are called "extrema"

- Absolute Maximum of f if $f(c) \geq f(x)$ for all x in the domain of f .
- Absolute Minimum of f if $f(c) \leq f(x)$ for all x in the domain of f .

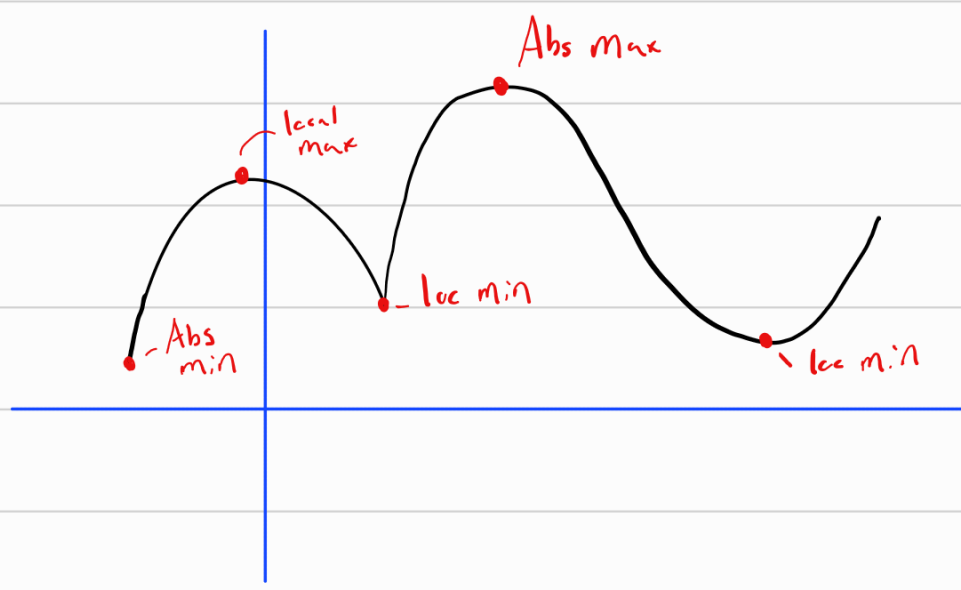
Def: For f , $f(c)$ is a

(relative) local Max if $f(c) \geq f(x)$ for x near c

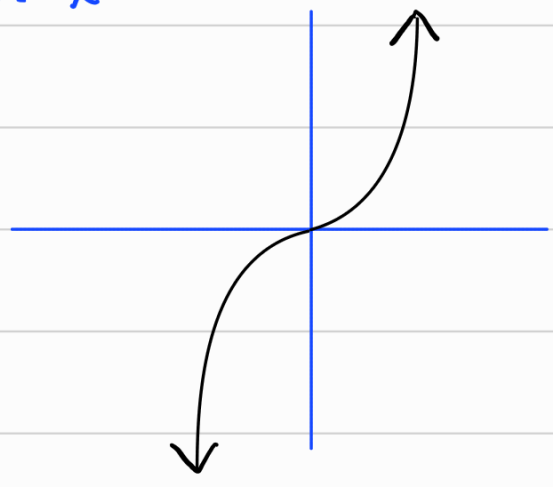
(relative) local Min if $f(c) \leq f(x)$ for x near c

Can't be an endpoint

Ex:



Ex: $f(x) = x^3$



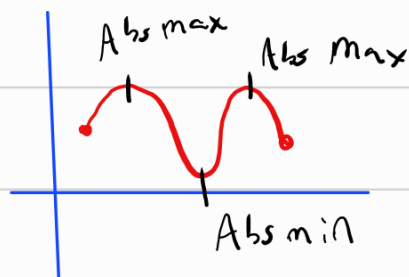
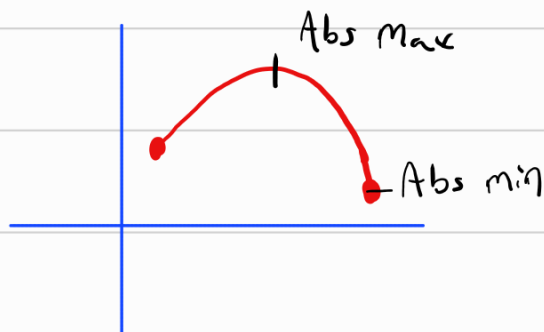
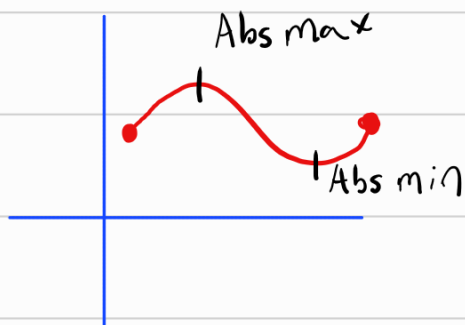
has no local or global extrema.

A good question to ask is when a function does attain a maximum value.

Thm (Extreme Value Theorem)

If f is continuous on a closed interval $[a, b]$, then f attains an absolute maximum $f(c)$ and absolute minimum $f(d)$ with $a \leq c, d \leq b$ ($c, d \in [a, b]$)

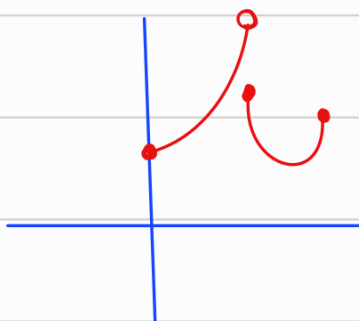
Ex:



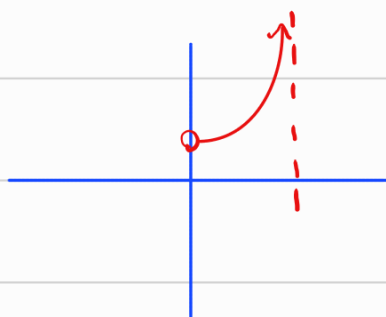
All these functions are continuous on a closed interval, so they all have absolute extrema, which may not be unique.

Note, the Theorem doesn't Apply to discontinuous functions.

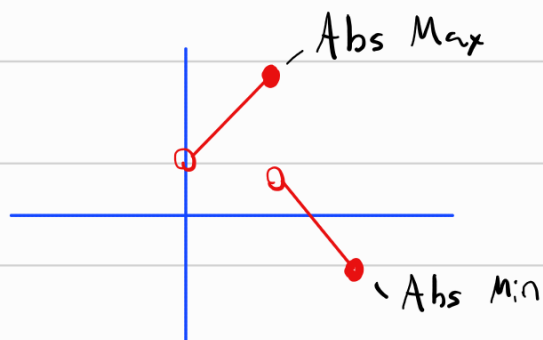
Ex:



This function has a local min, but not a local max.



No Absolute min nor max.



Finding maximums and minimums

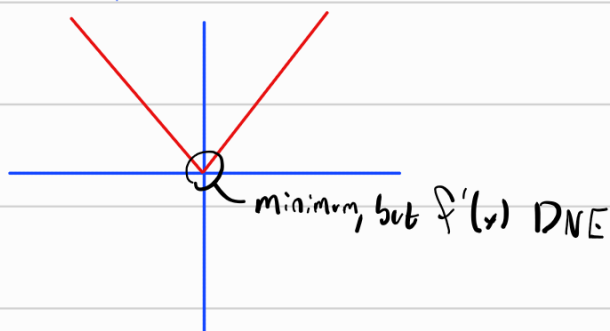
Thm (Fermat)

If f has a local maximum or minimum at c , and if $f'(c)$ exists, then $f'(c) = 0$

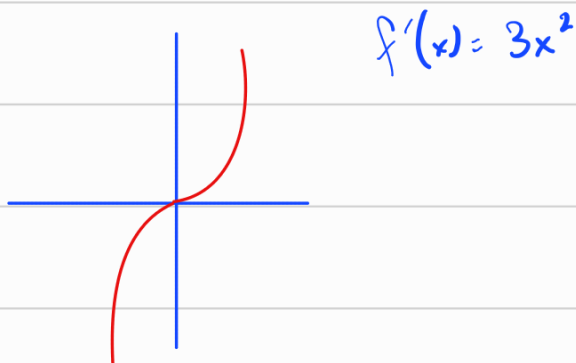
This theorem tells us that local extrema will only occur where $f'(c) = 0$ or $f'(c)$ DNE.

This does not mean if $f'(c) = 0$ or DNE that c is a local extrema, it just means c might be a local extrema.

Ex: $f(x) = |x|$



$f(x) = x^3$



Def: A Critical Number of a function f is a number c in the domain s.t. $f'(c) = 0$ or $f'(c)$ DNE

Ex: Find Critical Numbers of $f(x) = x^3 - 3x^2 + 1$

$$f'(x) = 3x^2 - 6x = 3x(x-2)$$

$$\text{if } f'(x) = 0 = 3x(x-2)$$

$$x = 0 \text{ or } x = 2$$

Critical Values are

$$x = 0, 2$$

Ex: Find Critical Values of $f(x) = x^{3/5}(4-x)$

$$f'(x) = \frac{3}{5}x^{-2/5}(4-x) - x^{3/5}$$

$$= \frac{12}{5}x^{-2/5} - \frac{3}{5}x^{3/5} - x^{3/5}$$

$$= \frac{12}{5x^{2/5}} - \frac{8x^{3/5}}{5}$$

$$= \frac{12}{5x^{2/5}} - \frac{8x}{5x^{2/5}} = \frac{12-8x}{5x^{2/5}}$$

$$f'(x) = 0 \text{ if } x = \frac{3}{2}$$

$$f'(x) \text{ DNE if } x = 0$$

Critical Values are $x = 0, \frac{3}{2}$

Now, an important take away,

If f has a local min/max at c , then c is a critical value.

This gives us a proper method for finding absolute extrema.

The Closed Interval Method

To find the maximum and minimum of a continuous function f on a closed interval $[a, b]$:

1. Find the values of f for all critical numbers in (a, b)
2. Find the value of f at the end points a and b .
3. The largest and smallest value in 1 and 2 are the maximum and minimum respectively

Ex: Find absolute max and min of

$$f(x) = x^3 - 3x^2 + 1 \quad \text{for} \quad -\frac{1}{2} \leq x \leq 4$$

Since f is continuous on $[-\frac{1}{2}, 4]$ we can use the closed interval method

1. Find value of f at critical numbers

Critical numbers:

$$f'(x) = 3x^2 - 6x = 3x(x-2)$$

$$\therefore \text{if } f'(x) = 0, \text{ then } x = 0, 2$$

Since $f'(x)$ is a polynomial, $f'(x)$ always exists.

Critical numbers are $x = 0, 2$

$$f(0) = \underline{1}$$

$$f(2) = 2^3 - 3(2)^2 + 1 = \underline{-3}$$

2. Value at end points

$$f(-\frac{1}{2}) = (-\frac{1}{2})^3 - 3(-\frac{1}{2})^2 + 1 = -\frac{1}{8} - \frac{6}{8} + \frac{8}{8} = \underline{\frac{1}{8}}$$

$$f(4) = 4^3 - 3(4)^2 + 1 = \underline{17}$$

3. Smallest Value above: $f(2) = -3$

Largest Value above: $f(4) = 17$

Absolute Max: $f(4) = 17$

Absolute Min: $f(2) = -3$

Ex: Find Absolute extrema of

$$f(x) = x - 2\sin(x) \quad \text{for} \quad 0 \leq x \leq 2\pi$$

1. Critical Values

$$\begin{aligned} f'(x) &= 1 - 2\cos(x) \\ \text{if } f'(x) &= 0 & \rightarrow & 1 - 2\cos(x) = 0 \\ & & & \cos(x) = \frac{1}{2} \\ & & & x = \frac{\pi}{3}, \frac{5\pi}{3} \end{aligned}$$

$$f\left(\frac{\pi}{3}\right) = \frac{\pi}{3} - 2\sin\left(\frac{\pi}{3}\right) = \frac{\pi}{3} - \sqrt{3}$$

$$f\left(\frac{5\pi}{3}\right) = \frac{5\pi}{3} - 2\sin\left(\frac{5\pi}{3}\right) = \frac{5\pi}{3} + \sqrt{3}$$

2. Endpoints

$$f(0) = 0 - 2\sin(0) = 0$$

$$f(2\pi) = 2\pi - 2\sin(2\pi) = 2\pi$$

3.

$$\text{Absolute Min: } f\left(\frac{\pi}{3}\right) = \frac{\pi}{3} - \sqrt{3} \approx -0.685$$

$$\text{Absolute Max: } f\left(\frac{5\pi}{3}\right) = \frac{5\pi}{3} + \sqrt{3} \approx 6.96$$

Ex: Find Absolute Extrema for

$$f(x) = x - \sqrt[3]{x} \quad \text{for} \quad -1 \leq x \leq 8$$

1. Critical Values

$$f'(x) = 1 + \frac{1}{3}x^{-2/3} = 1 + \frac{1}{3\sqrt[3]{x^2}}$$

$$\text{if } f'(x) = 0$$

$$1 - \frac{1}{3\sqrt[3]{x^2}} = 0$$

$$\Rightarrow x^{2/3} = \frac{1}{3} \quad \text{so} \quad x = \frac{\pm 1}{3\sqrt[3]{3}} = \underline{\underline{\pm \frac{\sqrt[3]{3}}{9}}}$$

$f'(x)$ DNE when $x=0$

$$f(0) = 0 - \sqrt[3]{0} = 0$$

$$\begin{aligned} f\left(\frac{\sqrt{3}}{9}\right) &= \frac{\sqrt{3}}{9} - \left(\frac{\sqrt{3}}{9}\right)^{1/3} = \frac{\sqrt{3}}{9} - \frac{3^{1/2}}{9^{1/3}} = \frac{3^{1/2}}{9^{1/3}} - \frac{3^{1/6}}{9^{1/3}} = \frac{3^{-5/6} - 3^{1/6}}{9^{1/3}} = \frac{3^{3/6} - 3^{4/6}}{9} \\ &= \frac{3^{1/2} - 3^{2/3}}{9} = \frac{\sqrt{3} - 3\sqrt{3}}{9} = -\frac{2\sqrt{3}}{9} \end{aligned}$$

$$f\left(-\frac{\sqrt{3}}{9}\right) = \frac{2\sqrt{3}}{9}$$

2. End Points

$$f(-1) = -1 - \sqrt[3]{-1} = 0$$

$$f(8) = 8 - \sqrt[3]{8} = 6$$

3. Abs Max: $f(8) = 6$

Abs Min: $f\left(\frac{\sqrt{3}}{9}\right) = -\frac{2\sqrt{3}}{9}$